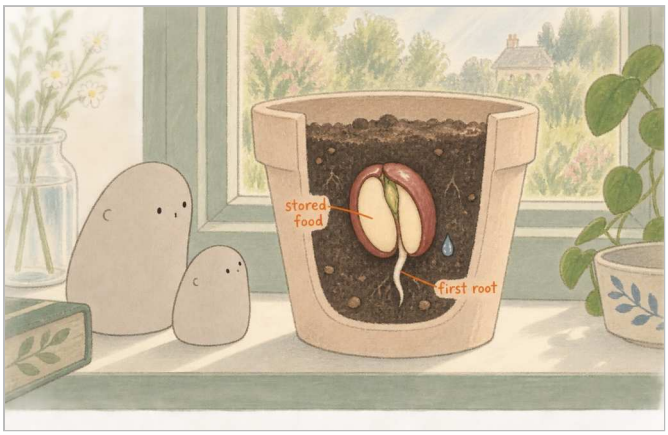
A bean becomes.

01 What is waking up inside this little bean?



An imagined window into the soil. The seed is enlarged to help us see.

Look closer. Look inside the seed coat. What might those two pale halves be keeping safe?

A little explanation. A tiny plant is already inside. The bean holds food to help it begin.

Try this. Curl one hand like a seed. Slowly uncurl a finger for its first root.

02 How does a root find its way when it cannot see?



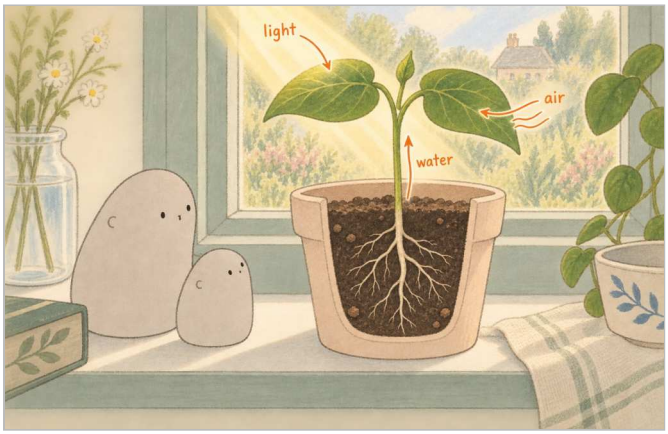
Roots and shoots respond to their surroundings. Plants do not need eyes.

Look closer. Follow the white root down. Now follow the little shoot up. What do you notice?

A little explanation. The root does not need eyes. It responds to gravity, which helps it grow down.

Try this. Follow the two directions in the air: root down, shoot up.

03 What can a leaf do that a seed cannot?



Leaves make new food using light, water, and carbon dioxide from the air.

Look closer. Find the three notes around the leaves. Where does each arrow lead?

A little explanation. Green leaves use light, water, and part of the air to make new food. The seed began with food stored inside.

Try this. Follow a path from the roots to a leaf. Then point to the light.

04 How could we notice a change too slow to watch?



Compare from the soil level. This painted stick is not a scale ruler.

Look closer. Find the earlier mark and today's tip. What could a drawing help us remember?

A little explanation. We can compare notes from different days. Growth can happen too slowly for our eyes to catch it happening.

Try this. Hold a finger at the earlier mark. With your other hand, show the space up to today's tip.